

**Hyunseung Hwang**  
email: aguno@kais.ac.kr  
mobile number: 010-4003-7805  
Website: <https://aguno.github.io>

## **Professional Summary**

---

Machine learning researcher specializing in AI interpretability, model robustness, and alignment. I stress-test the explanation methods practitioners rely on to audit high-stakes models, exposing their hidden vulnerabilities and engineering more transparent, robust alternatives. My work spans from developing explainability-first clustering frameworks to demonstrating how widely used tools like SHAP can be highly sensitive to data-engineering choices. I aim to build trustworthy AI systems where model risk is strictly managed, ensuring that automated decisions in critical domains—from AI alignment to financial services—are safe, accountable, and highly confident.

## **Education**

---

Ph.D. School of Electrical Engineering, KAIST, Sep 2021 - Aug 2026  
M.S., School of Electrical Engineering, KAIST, Sep 2019 -Aug 2021  
B.S., School of Computing, KAIST, Sep 2013 -Aug 2019  
Langley High School, Fairfax County, VA Sep 2010- Aug 2013

## **WORK EXPERIENCE**

---

**New York University (NYU), Center for Responsible AI** | New York, NY *Incoming Postdoctoral Researcher* (Sep 2026 – Present)

- Will conduct advanced research on the interpretability and safety of machine learning models under the supervision of Prof. Julia Stoyanovich.

**Global Frontier Lab** | Daejeon, South Korea *Research Collaborator* (Fall 2025)

- Collaborated with the NYU Center for Responsible AI to investigate the sensitivity and robustness of AI explainability techniques.
- Developed methodologies to stress-test explanation models, ensuring safe, confident, and defensible outputs for end-users and stakeholders.

**Naver Corporation** | Seongnam, South Korea *Research Intern* (Summer 2019)

- Implemented large-scale community detection algorithms (Girvan-Newman) using PySpark.
- Clustered massive unsupervised datasets to optimize and enhance the video recommendation engine.

**Kiswe Mobile** | *Software Engineering Intern* (Summer 2018)

- Engineered and shipped core Android features for a live video-streaming service.
- Upgraded the application architecture to meet and exceed Android Oreo (8.0+) performance and security requirements.

## **Published Papers**

---

SHAP-based Explanations are Sensitive to Feature Representation

- Accepted at ACM FAccT 2025

XClusters: explainability-first clustering

- Accepted at AAAI Conference on Artificial Intelligence 2025

Open-World COVID-19 Data Visualization

- Accepted at VLDB Workshop on Data Management and Healthcare 2021